

Introduction to Signals and Systems

- Signal - Dependent variable or function of one or more independent variables.

$$F(x_1, x_2, \dots, x_n)$$

↓
signal.

independent variables

- For example
1. AC (alternating current) is a signal - (current changes with time)
 2. DC (Direct Current) not a signal.

Types of signal

Single Variable signal

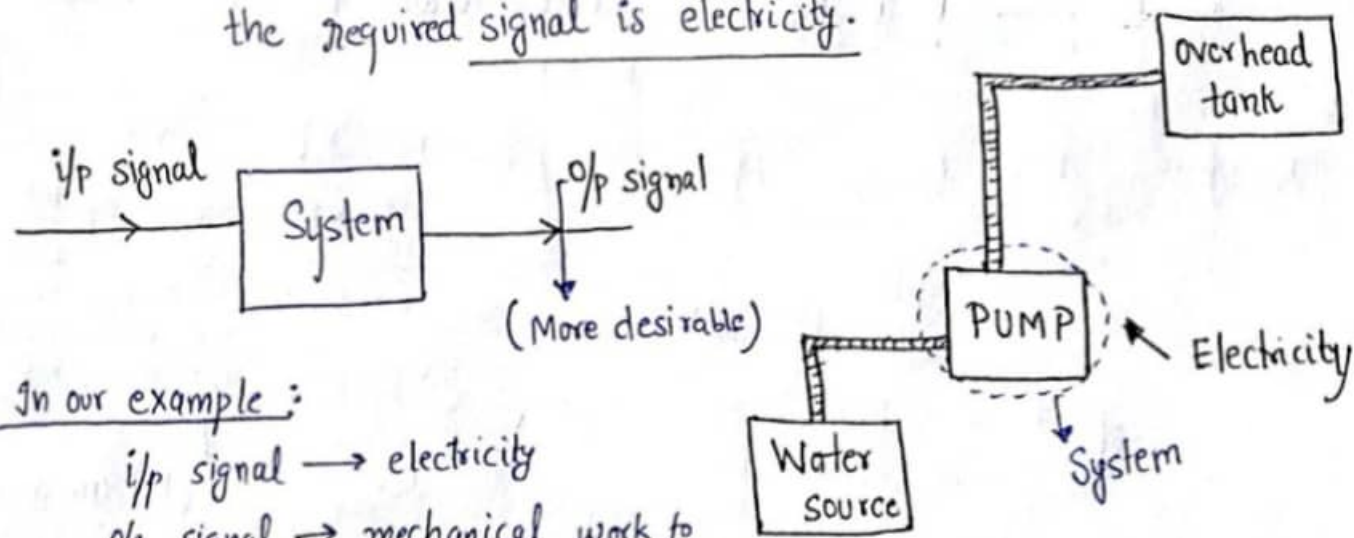
- ↳ function of one variable
- ↳ Ex: $f(t)$, $g(x)$, etc.

Multivariable signal

- ↳ function of ^{more than} one variable.
- ↳ Ex. $f(t_1, t_2)$, $g(x_1, x_2)$, etc.

System: The meaningful interconnection of physical devices and components is called as system.

Example: Pump is our system. But a system alone cannot perform a task, hence it requires a signal. In this case the required signal is electricity.



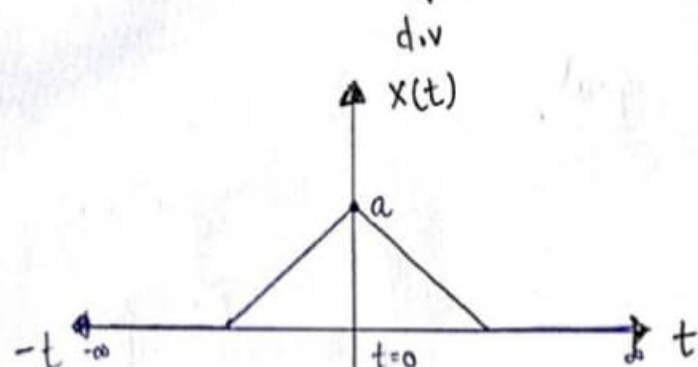
Remember : It is not necessary that o/p signal is more desirable.
I/p signal and o/p signal depend on the system.

→ Continuous and Discrete Time signals

Continuous Time Signals (CTS)

1. Specified for every value of time (t).

2. Representation : $x(t)$



y-axis → always for signal/function

x-axis → always for Independent variable (i.v)

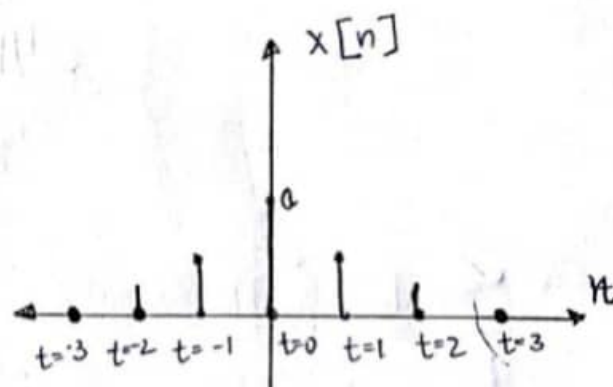
a → amplitude at $t=0$.

Discrete time signals (DTS)

1. Specified at discrete time intervals.

2. Representation : $x[n]$

$n \rightarrow \text{integer}$

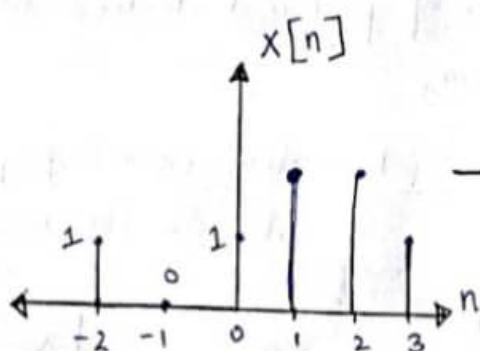


$$\Delta t = t_1 - t_0, t_2 - t_1$$

If $\Delta t = t_1 - t_0 = t_2 - t_1 = \dots = t_n - t_{n-1}$

DTS → is uniformly sampled
else non uniformly sampled.

Example :



→ Given plot of DTS.

→ Using this example we achieve another way of representation for DTS.

Step 1

$$x\{n\} = \{1, 0, 1, 2, 2, 1\}$$

Amplitude values of signal.

(But we don't know for what values of 'n' these amplitude values correspond)

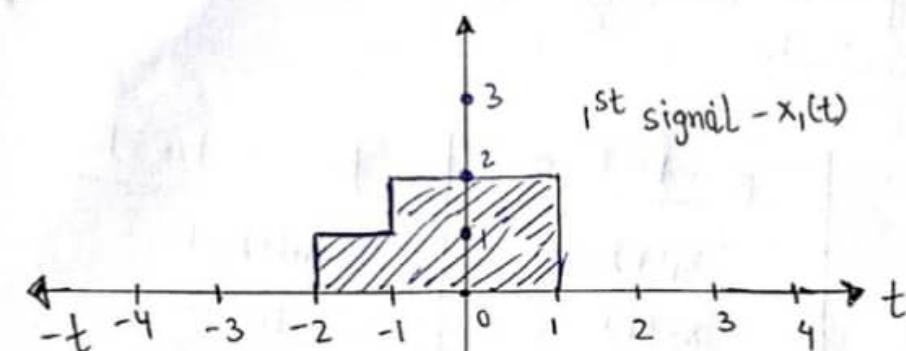
Step 2 Put $x[n] : n=0$

At $x[0]$, the amplitude value = 1.

$$x\{n\} = \{1, 0, 1, 2, 2, 1\}$$

(Put an arrow)

→ Addition of CTS



$$\text{At } t=2, (1 < t \leq 2)$$

$$x_1(t) = 0$$

$$x_2(t) = 2$$

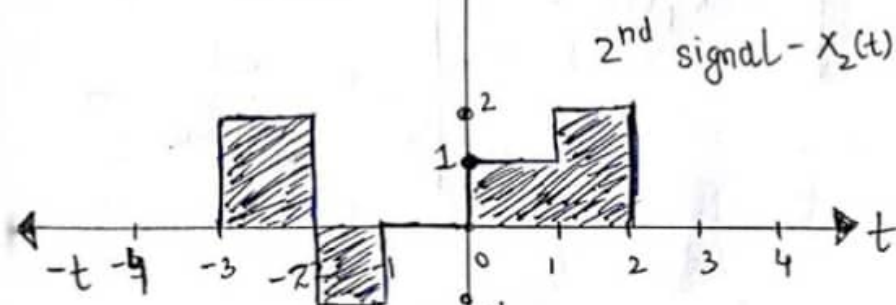
$$\therefore x_3(t) = 2$$

$$\text{for } 0 < t \leq 1$$

$$x_1(t) = 2$$

$$x_2(t) = 1$$

$$\therefore x_3(t) = 3$$



$$x_3(t) = x_1(t) + x_2(t)$$

$$\text{for } -1 < t \leq 0$$

$$x_1(t) = 2$$

$$x_2(t) = 0$$

$$\therefore x_3(t) = 2$$

Addition
of CTS



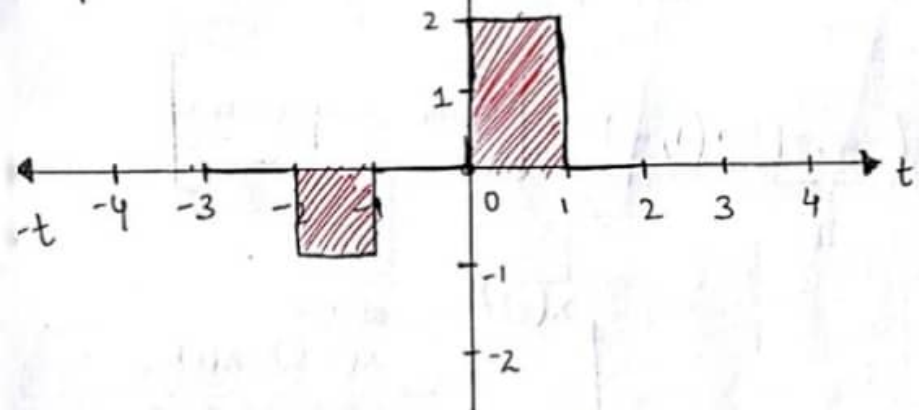
$$\text{for } -2 < t \leq -1$$

$$x_1(t) = 1$$

$$x_2(t) = -1$$

$$\therefore x_3(t) = 0$$

Multiplication
of CTS



$$\text{for } -3 < t \leq -2$$

$$x_1(t) = 0$$

$$x_2(t) = 2$$

$$\therefore x_3(t) = 2$$

• Multiplication of CTS

→ Multiply amplitude values of both the given signal in every time interval and plot accordingly.

for $1 < t \leq 2$	for $0 < t \leq 1$	for $-1 < t \leq 0$	for $-2 < t \leq -1$
$x_1(t) = 0$	$x_1(t) = 2$	$x_1(t) = 2$	$x_1(t) = 1$
$x_2(t) = 2$	$x_2(t) = 1$	$x_2(t) = 0$	$x_2(t) = -1$
$x_4(t) = 0$	$x_4(t) = 2$	$x_4(t) = 0$	$x_4(t) = -1$
			for $-3 \leq t \leq -2$
			$x_4(t) = 0$

→ Scaling

→ Time Scaling (Amplitude remains same)

→ Amplitude scaling. (Time remains same)

1. Time Scaling of CTS : The compression or expansion of a signal in time

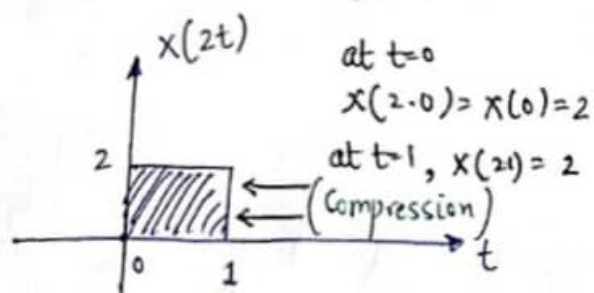
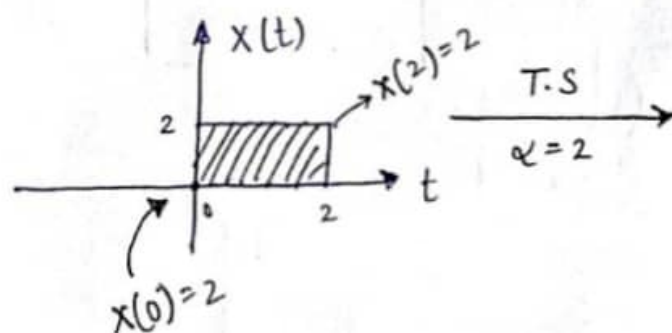
$$x(t) \xrightarrow{T.S} y(t) = x(\alpha t), \alpha \neq 0$$

↓
(original signal)

- In time scaling we multiply time (t) by a non-zero number to achieve $y(t)$.

Case I : $|\alpha| > 1$, $\alpha \in (-\infty, -1) \cup (1, \infty)$ [Time compression]

Example : $x(t) = 2, 0 \leq t \leq 2$



$$x(2t) = 2$$

$$0 \leq t \leq 1 \Rightarrow \frac{0}{\alpha} \leq \frac{2t}{\alpha} \leq \frac{2}{\alpha}$$

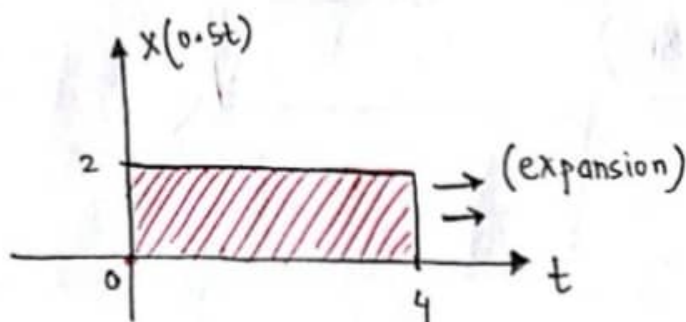
Remember : 1. Amplitude remains same

2. $\frac{\alpha t}{\alpha}$

Case II : $|\alpha| < 1$ and $\alpha \neq 0$ $\alpha \in (-1, 0) \cup (0, 1)$ { Expansion }

Example : Let us consider previous problem but $\alpha = 0.5$

1. Amplitude remains same.
2. We divide αt axis by α , which gives x -axis as ' t '
3. We divide upper & lower ends of $x(t)$ $\left[\frac{2}{0.5} = 4, \frac{0}{0.5} = 0 \right]$



2. Amplitude scaling of CTS

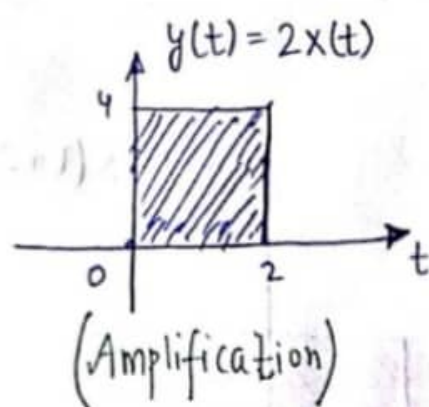
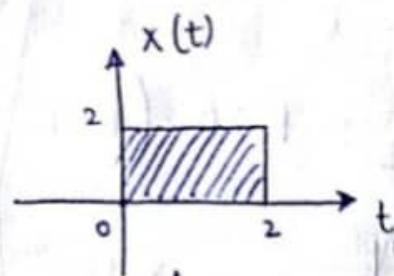
↳ Multiply amplitude of signal by a real no. ' β '

$$\text{↳ } x(t) \xrightarrow[\beta]{A \cdot s} y(t) = \beta x(t)$$

Case I : $|\beta| > 1$, $\beta \in (-\infty, -1) \cup (1, \infty)$ [Amplification]

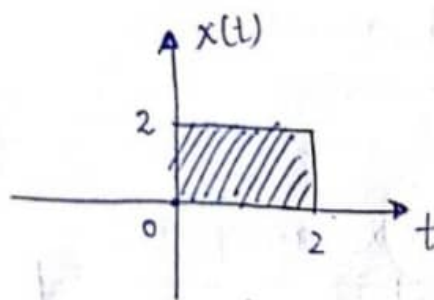
Example :

$$x(t) = \begin{cases} 0 & t < 0 \\ 2 & 0 \leq t \leq 2 \\ 0 & t > 2 \end{cases}$$

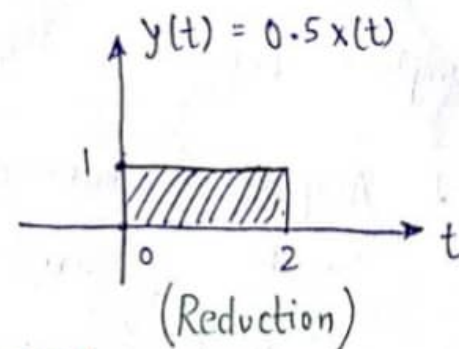


$A \cdot s$
($\beta = 2$) > 1

Case II $|\beta| < 1$ $\beta \in (-1, 0) \cup (0, 1)$ [Reduction]



$A.S$
 $\beta = 0.5$



NOTE : Same rules can be applied on Discrete Time signals. (DTS)

→ Shifting

↳ Time Shifting

↳ Amplitude Shifting.

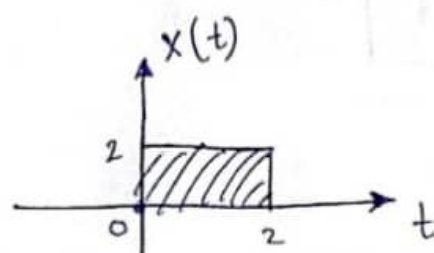
• Time Shifting of CTS

$$x(t) \xrightarrow[k \downarrow \text{Constant (in seconds)}]{T.S} y(t) = x(t+k)$$

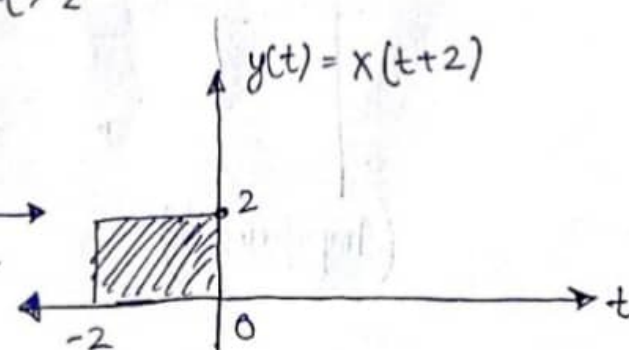
Case I : $k > 0$ (k is +ve)

Example :

$$x(t) = \begin{cases} 0 & t < 0 \\ 2 & 0 \leq t \leq 2 \\ 0 & t > 2 \end{cases}$$



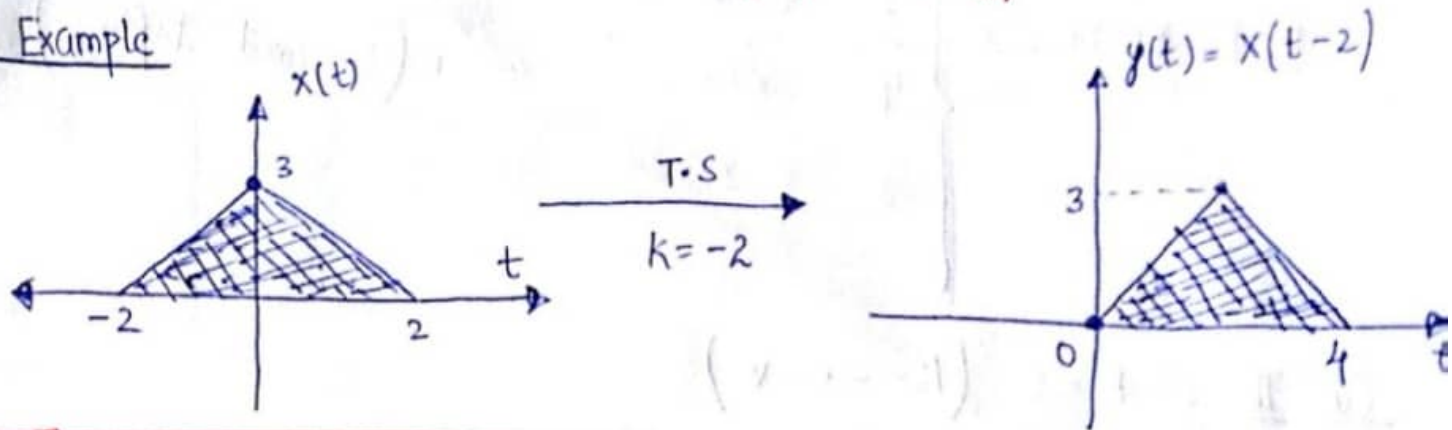
$T.S$
 $k = +2$



* $y(t) = x(t - (-2))$ (Left shifting or time advance)

Case II : $k < 0$ ($k \rightarrow -ve$) (Right Shifting)

Example



$$y(t) = x(t-2) = x(t - (+2)) \rightarrow \text{Right shifting or Time delay}$$

Remember : In real time we cannot advance a signal.
Only its recorded version can be shifted.

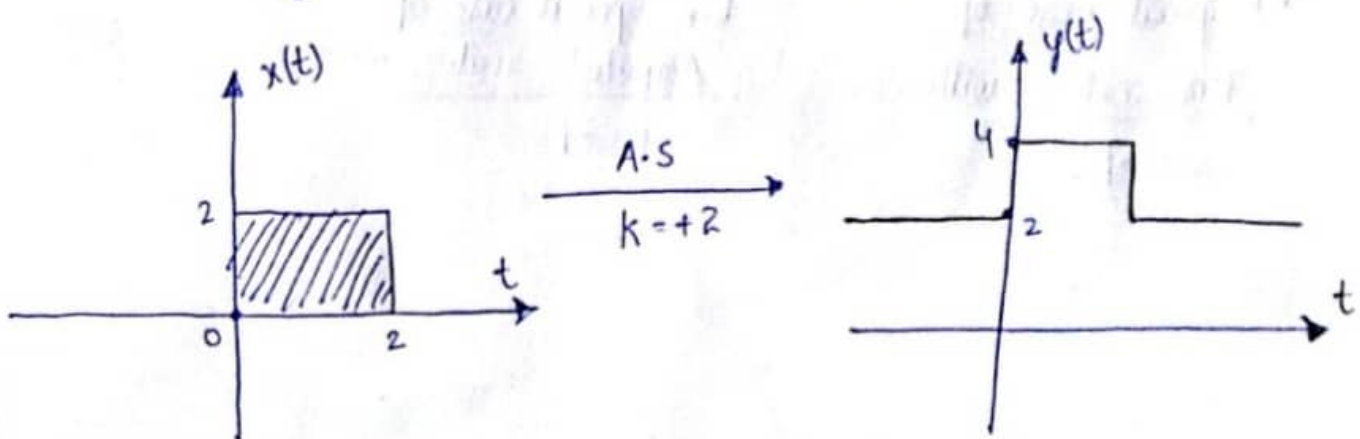
Amplitude Shifting of CTS

$$x(t) \xrightarrow[k]{A.S} y(t) = x(t) + k$$

Case I : $k > 0$ ($k \rightarrow +ve$) $\rightarrow$ (Upward Shifting)

$$x(t) = \begin{cases} 0 & t < 0 \\ 2 & 0 \leq t \leq 2 \\ 0 & t > 2 \end{cases} \quad \text{and let } k = +2$$

and $y(t) = x(t) + 2$



$$y(t) = x(t) + 2 = \begin{cases} 2 & t < 0 \\ 4 & 0 \leq t \leq 2 \\ 2 & t > 2 \end{cases}$$

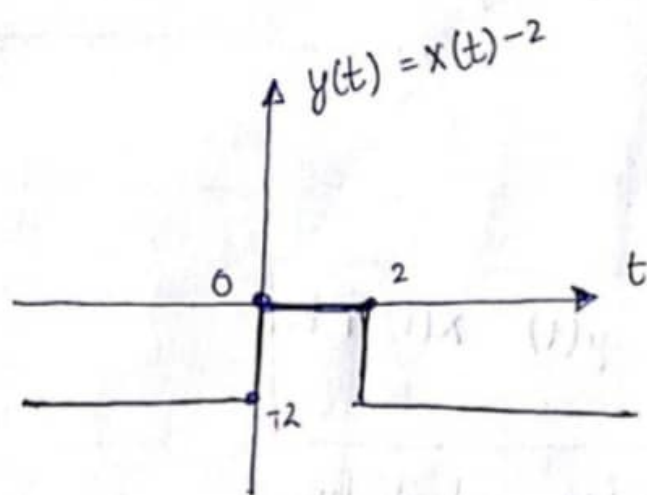
→ (Upward shifting)

Case II : $k < 0$ ($k \rightarrow -ve$)

$$x(t) = \begin{cases} 0 & t < 0 \\ 2 & 0 \leq t \leq 2 \\ 0 & t > 2 \end{cases}$$

let $k = -2$

$$y(t) = \begin{cases} -2 & t < 0 \\ 0 & 0 \leq t \leq 2 \\ -2 & t > 2 \end{cases}$$



→ (Downward Shifting)

• Reversal of CTS

Time reversal

↳ Special case of
Time scaling with
 $\alpha = -1$.

Amplitude Reversal

↳ Special case of
Amplitude scaling with
 $\beta = -1$

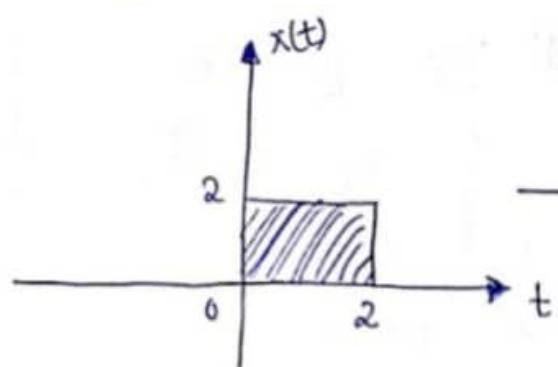
• Time Reversal of CTS (Folding or Reflection)

↳ special case of Time Scaling with $\alpha = -1$

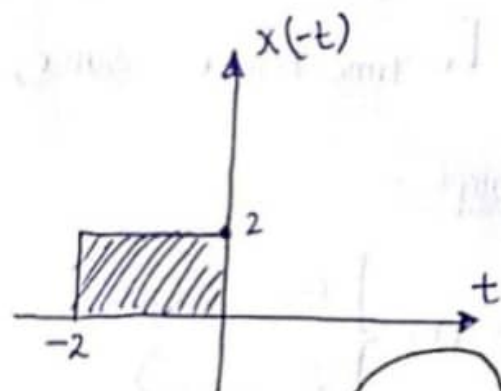
$$\text{↳ } x(t) \xrightarrow{T.S} y(t) = x(\alpha t)$$

$$\text{↳ } \alpha = -1, \quad x(t) \xrightarrow{T.R} y(t) = x(-t)$$

Example :



$\xrightarrow[\alpha = -1]{T.R}$



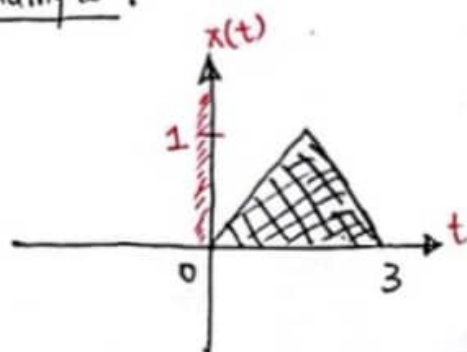
$$x(t) = \begin{cases} 0 & t < 0 \\ 2 & 0 \leq t \leq 2 \\ 0 & t > 2 \end{cases}$$

$$x(-t) = \begin{cases} 0 & -t < 0 & t > 0 \\ 2 & 0 \leq -t \leq 2 & 0 \geq t \geq -2 \\ 0 & -t > 2 & t < -2 \end{cases}$$

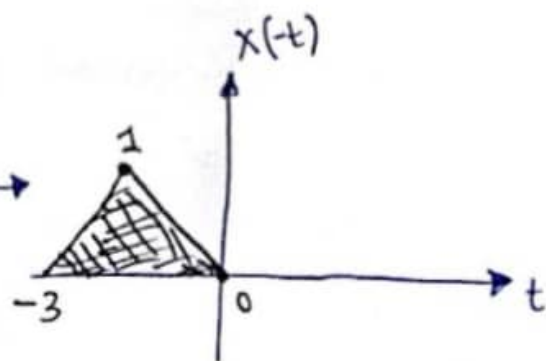
* We divide the inequality in $x(-t)$ by -1 , also the signs interchange.

* In time reversal, the new signal is mirror image of original signal. (or by folding the original signal across y-axis)

Example :



$\xrightarrow[\alpha = -1]{T.R}$



• Amplitude Reversal

↳ Special case of Amplitude scaling with $\beta = -1$.

$$\text{↳ } x(t) \xrightarrow{A \cdot s} y(t) = \beta x(t)$$

↳ In A.R, $\beta = -1$

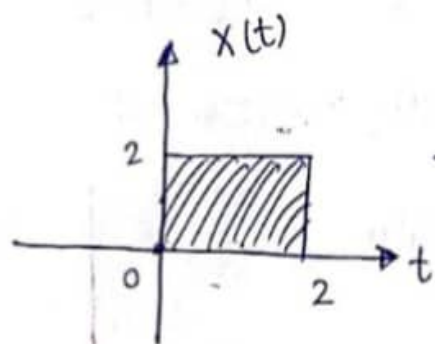
$$\text{↳ } \boxed{x(t) \xrightarrow{A \cdot R} y(t) = -x(t)}$$

↳ Time Remains same, amplitude changes.

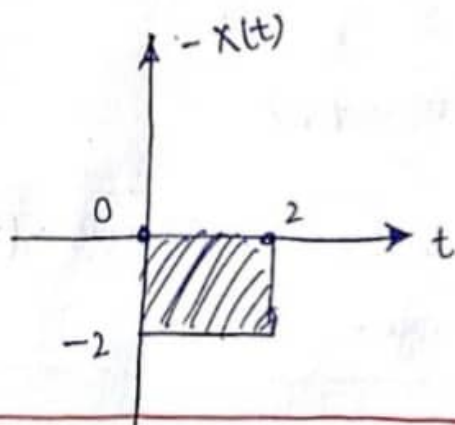
Example :

$$x(t) = \begin{cases} 0 & t < 0 \\ 2 & 0 \leq t \leq 2 \\ 0 & t > 2 \end{cases}$$

$$-x(t) = \begin{cases} 0 & t < 0 \\ -2 & 0 \leq t \leq 2 \\ 0 & t > 2 \end{cases}$$



$\xrightarrow{A \cdot R}$



• In Amplitude reversal, the waveform of new signal is mirror image of original waveform of signal, folded about x-axis.

• Multiple transformations of CIS

1. Scaling
2. Shifting
3. Reversal.

Example : $x(t)$ $\rightarrow$ original signal
and plot $y(t) = A x\left(\frac{-t-t_0}{a}\right)$

Solⁿ : Step.1 $\rightarrow$ Divide $-t/a$ and $-t_0/a$ separately.

$$y(t) = A x\left(\frac{-t}{a} - \frac{t_0}{a}\right)$$

Step.2 $\rightarrow$ Given in the problem statement. (Plot $x(t)$)

Step.3 $\rightarrow$ Plot $Ax(t)$ [Amplitude scaling]

Step.4 $\rightarrow$ Plot $Ax(t)$ [Time reversal]

Step.5 $\rightarrow$ Plot $Ax(-t - t_0/a)$ [Time shifting]

Step.6 $\rightarrow$ Plot $Ax\left(\frac{-t}{a} - \frac{t_0}{a}\right)$ [Time scaling]

This order must not change.

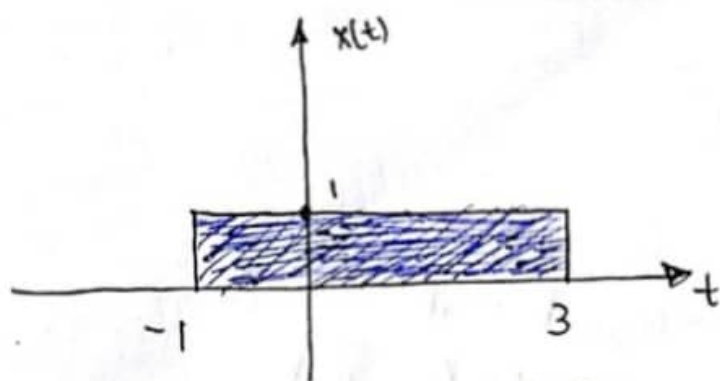
* Follow the steps in sequence

Priority order :

1. Time Reversal
2. Time Shifting
3. Time Scaling

$\rightarrow$ Amplitude scaling can be done at any time.

Ques: Plot the signal $y(t) = 3x(2t+3)$, where $x(t)$ is shown in figure.

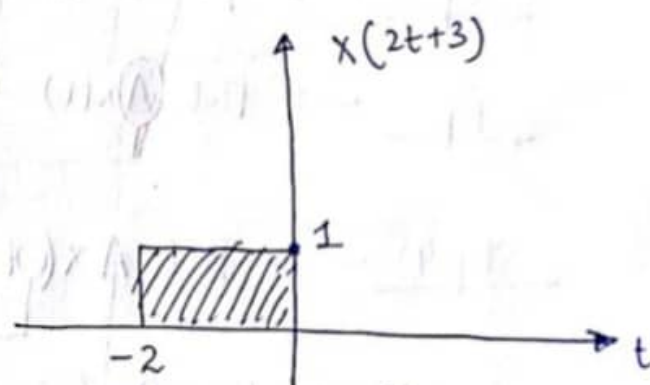
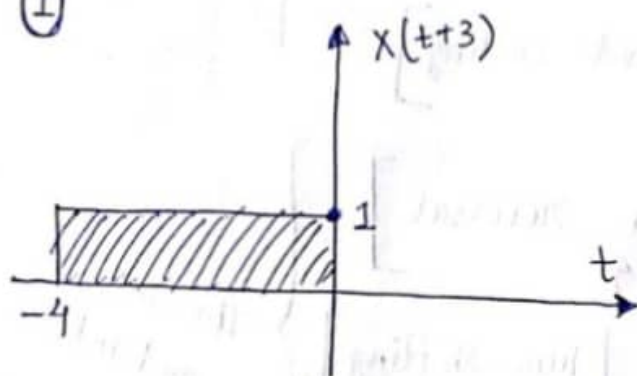


Sol: (i) Since $y(t)$ is not having negative ' t ' so no need to perform time reversal.

(ii) Time Shifting

$$x(t) \xrightarrow[k=3]{T.S} x(t+3) = x(t - (-3))$$

①



$$(iii) \quad x(t+3) \xrightarrow[\alpha=2]{T.S} x(2t+3) \quad \left[\begin{array}{l} \text{Compressed} \\ \text{by factor of } 2 \end{array} \right]$$

(iv) Amplitude scaling

$$x(2t+3) \xrightarrow[\beta=3]{A.S} 3x(2t+3)$$

